SIT720 - 10.2 Credit Task

Neural Networks for Image Classification - Fashion-MNIST

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Status of this report

This companion PDF is a pre-run version. The supplied Jupyter notebook contains the complete implementation for all required experiments and automatically generates a final PDF report from the actual executed results. Numerical model-performance values are deliberately not invented in this pre-run document.

Assessment coverage

The notebook loads Fashion-MNIST; reports dataset size, image dimensions and classes; visualises representative examples; trains a one-hidden-layer baseline MLP; investigates depths of 2, 4, 6, 8 and 10 with 100 neurons per layer; investigates widths of 20, 50, 100 and 200 using two hidden layers; compares depth and width with accuracy and training time; tunes activation and L2 regularisation; performs confusion-matrix and prediction error analysis; and gives an evidence-based final MLP recommendation.

How to obtain the final submission PDF

Open the notebook in Google Colab and choose Runtime > Run all. Fashion-MNIST will be downloaded automatically. The final cell creates SIT720_10_2C_Final_Report_Zenith_Anthony.pdf using the actual results from that run. The report therefore contains genuine model accuracies, training times, hyperparameter-selection evidence, and the most frequent misclassification pair.

Academic integrity note

The task brief permits GenAI for planning, brainstorming and editing, but requires the student to critically review, understand and appropriately modify AI-assisted material. Review every section, make sure you understand the MLP design and results, and retain the GenAI acknowledgement.